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Assignment 1: Automated Container Build Pipeline

Objective: Grasp the fundamentals of containerisation and automated workflows.

Scenario: Package a simple application into a container and set up an automated and pipeline that builds and stores the container image every time code is committed.

Tasks:

- **App Setup:** Create a basic "Hello World" web application (Node.js/Python).
- **Containerize:** Write a Dockerfile to package and run the app.
- **Automation:** Create a GitHub Actions workflow that triggers on push to main, builds the image, and pushes it to Docker Hub .

Deliverables: Repo link with code/Dockerfile, and a link to the successful GitHub Actions run along with screenshots.

Assignment 2 : Foundational Cloud Provisioning

Objective: Provision and manage cloud infrastructure using declarative code.

Scenario: Provision a secure network environment and a basic compute server on AWS entirely through code.

Tasks:

- **Networking: Write main.tf to create a VPC, public subnet, and Internet**
- **Gateway.**
- **Compute: Provision a t2.micro EC2 instance with a security group allowing**
- **HTTP/SSH.**

- **Outputs: Use Terraform outputs to display the public IP.**
- **Lifecycle: Execute init, plan, apply, and finally destroy.**
- **Deliverables: The .tf files and screenshots of plan/apply/destroy terminal outputs.**